

ANFIS-Based Cardiac Risk Classification: Comparing Neuro-Fuzzy Systems and MLP for Accuracy and Interpretability

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Abstract—This paper investigates the tradeoff between predictive accuracy and interpretability in machine learning models applied to cardiac risk classification. We propose an ANFIS (Adaptive Neuro-Fuzzy Inference System) with Takagi-Sugeno zero-order inference and mixed fuzzification—Gaussian membership functions for numerical variables and smoothed one-hot encoding for categorical variables—whose rule pool is generated via K-Means clustering. Experiments on the Heart Failure Prediction Dataset (918 patients, 11 features) show that ANFIS achieves AUC-ROC 0.912, accuracy 84.2%, and F1-Score 85.9%, outperforming the MLP baseline (AUC 0.901; accuracy 81.5%; F1 82.5%) with only 228 trainable parameters versus 2,881 in the MLP. Beyond superior performance, ANFIS produces 10 auditable IF-THEN linguistic rules consistent with the ischemia literature.

Keywords—ANFIS; Fuzzy Logic; Neural Networks; Heart Disease; Interpretability; Takagi-Sugeno; K-Means

I. INTRODUCTION

Clinical decision support systems based on machine learning achieve high accuracy in diagnostic tasks, yet frequently operate as black boxes, producing predictions without intelligible justification [1]. In the medical context, decision traceability is both an ethical and regulatory requirement: a specialist must understand *why* a patient was flagged as high-risk in order to validate or challenge the recommendation.

Neuro-Fuzzy systems—ANFIS in particular—combine the learning capability of neural networks with the interpretability of fuzzy inference systems [2]. Linguistic rules of the form IF *antecedent* THEN *consequent* allow clinicians and auditors to inspect decision criteria without requiring machine learning expertise.

This paper contributes: (i) an ANFIS architecture with mixed fuzzification to handle numerical and categorical features in a unified pipeline; (ii) a K-Means-based rule pool generation method that scales to datasets with multiple categorical variables; and (iii) a quantitative comparison with an MLP on the Heart Failure Prediction Dataset [3], showing that ANFIS outperforms the MLP on all metrics with 13× fewer parameters.

II. DATASET AND PREPROCESSING

A. Heart Failure Prediction Dataset

The dataset combines records from five clinical sources, totalling 918 adult patients. The target variable **HeartDisease** is binary (0=healthy, 1=disease) with ~55% positive prevalence. The train/test split is 80/20 with class stratification.

The 11 features comprise 5 numerical and 6 categorical variables (Table I).

Feature	Type	Description
Age	Numerical	Patient age (years)
RestingBP	Numerical	Resting blood pressure (mmHg); zeros replaced by median
Cholesterol	Numerical	Serum cholesterol (mg/dL); zeros replaced by median
MaxHR	Numerical	Maximum heart rate achieved (bpm)
Oldpeak	Numerical	ST segment depression induced by exercise (mm)
Sex	Categorical	M / F
ChestPainType	Categorical	ATA / NAP / ASY / TA
FastingBS	Categorical	Fasting blood sugar > 120 mg/dL: 1=yes, 0=no
RestingECG	Categorical	Normal / ST / LVH
ExerciseAngina	Categorical	Exercise-induced angina: Y / N
ST_Slope	Categorical	Peak-exercise ST slope: Up / Flat / Down

TABELA I. Dataset features.

B. Preprocessing

Zero values in **RestingBP** and **Cholesterol** (clinically implausible) are replaced by the column median. Categorical features are integer-encoded via *pd.Categorical.codes*. Normalization uses a custom **PartialScaler**: z-score only on the 5 numerical features, preserving integer codes for categoricals—essential so that *MixedFuzzyLayer* can cast categorical indices without corruption.

III. ANFIS ARCHITECTURE

The proposed ANFIS follows the zero-order Takagi-Sugeno paradigm with three trainable layers. The data flow is:

$$X [B,11] \rightarrow \text{MixedFuzzyLayer} \rightarrow [B,11,M] \rightarrow \text{ClusterRuleLayer} \rightarrow [B,R] \rightarrow \text{DefuzzyLayer} \rightarrow [B,1] \rightarrow \text{sigmoid} \rightarrow P(\text{disease})$$

where B is the batch size, M the number of MFs per numerical feature, and R the number of rules in the pool.

A. MixedFuzzyLayer

Numerical features — trainable Gaussian MFs with center *c* and spread *s*, optimized by backpropagation:

$$\mu(x) = \exp(-0.5 * ((x - c) / s)^2)$$

Spreads are clamped at $s \geq 1e-6$ and initialized as $s = |N(0,1)| + 0.1$ to guarantee positivity.

Categorical features — smoothed one-hot encoding, no trainable parameters. For a variable with n categories:

$$\mu_j(cat) = 0.95 \text{ if } j = cat, \text{ else } 0.05 / (n - 1)$$

This provides a non-zero gradient to non-matching positions, preventing rule collapse during training.

B. ClusterRuleLayer

The firing strength of rule k is the product T-norm over all features:

$$w_k = \text{product}_i \mu_{\{i, \text{combo}[k,i]\}}(x_i)$$

where $\text{combo}[k,i]$ is the MF index of rule k in feature i . The combination table is stored as a register_buffer (automatic GPU migration). No trainable parameters.

C. DefuzzyLayer

Normalized Takagi-Sugeno defuzzification:

$$w_{\sim k} = w_k / (\sum_j w_j + \text{eps})$$

$$y_{\text{hat}} = \sum_k w_{\sim k} * c_k$$

The consequents c_k are weights of $\text{nn.Linear}(R \rightarrow 1)$. Normalization keeps output scale independent of how many rules fire simultaneously, preserving the linguistic interpretation of each c_k .

D. Loss Function

The loss combines binary cross-entropy with a center-separation penalty:

$$L = \text{BCE}(\text{sigma}(y_{\text{hat}}), y) + \text{lambda} * \sum_{\{i < j\}} \text{relu}(s_i + s_j - |c_i - c_j|)$$

The penalty ($\text{lambda}=0.1$) is zero when adjacent MF centers are at least $s_i + s_j$ apart, and grows linearly otherwise, preventing a narrow MF from collapsing inside a wider one.

IV. RULE POOL GENERATION VIA K-MEANS

The number of antecedent combinations grows exponentially: $n_{\text{rules}} = M^{n_{\text{features}}}$. For 11 features with $M=3$ this yields 177,147 rules—infeasible. The following five-step pipeline generates a manageable rule pool before training:

Step 1 — Percentile peaks (mapping artifact only): For each numerical feature, n_{mfs} uniformly-spaced percentiles of the training distribution (P25, P50, P75 for $M=3$) serve as auxiliary triangular MF peaks for cluster-to-index mapping. These are NOT the ANFIS Gaussian MFs.

Step 2 — K-Means: 200 clusters on the training feature space identify the highest-density regions.

Step 3 — Dominant MF mapping: For each cluster center, the dominant MF index is: numerical $\rightarrow \text{argmax}$ of triangular MF at peaks; categorical $\rightarrow \text{round}(\text{center}) \rightarrow$ integer code.

Step 4 — Deduplication: Clusters with the same dominant-MF vector collapse into one rule. 200 clusters $\rightarrow$ 197 unique combinations in this run.

Step 5 — Initialization: ANFIS Gaussian centers are initialized at percentiles of the K-Means cluster centers per feature, providing an informed starting point for end-to-end training.

V. EXPERIMENTAL SETUP

Hyperparameters are listed in Table II. Both models use the Adam optimizer with a plateau learning-rate scheduler and early stopping (patience 20 epochs).

Parameter	ANFIS	MLP
Epochs	200	200
Batch size	32	32
Learning rate	1e-3	1e-3
Architecture	3 MFs, 197 rules	[64, 32]
Parameters	228	2,881
Regularization	Sep. penalty $\lambda=0.1$	—
Seed	42	42

TABELA II. Experimental hyperparameters.

Evaluation metrics on the held-out test set (20%): AUC-ROC (threshold-independent), accuracy (threshold 0.5), and F1-Score (harmonic mean of precision and recall). All randomness is seeded at 42 across numpy, torch, and sklearn.

VI. RESULTS AND DISCUSSION

A. Performance Metrics

Table III summarizes the quantitative results. ANFIS outperforms the MLP on all three metrics: AUC-ROC (+0.011), Accuracy (+2.7 p.p.), F1 (+3.4 p.p.).

Model	Param s.	Train (s)	Accurac y	F1-Score	AUC-RO C
ANFIS	228	14.9	84.2%	85.9%	0.912
MLP	2,881	4.8	81.5%	82.5%	0.901

TABELA III. Comparative results — Heart Failure dataset, 80/20 split.

Parametric efficiency is notable: 228 vs. 2,881 parameters (13× fewer), reducing overfitting risk. The trade-off is ~3× slower training (14.9 s vs. 4.8 s), acceptable for offline diagnostic use.

B. Learned Membership Functions

Fig. 1 shows post-training Gaussian MFs for all 5 numerical features. Learned centers reveal clinically coherent thresholds: in **Age**, the three MFs separate young (~40 yr), middle-aged (~55 yr), and elderly (~65 yr) patients; in **MaxHR**, the "low" MF concentrates near ~120 bpm; in **Oldpeak**, the "high" MF captures depressions above ~1.5 mm—the established ischemia diagnostic threshold.

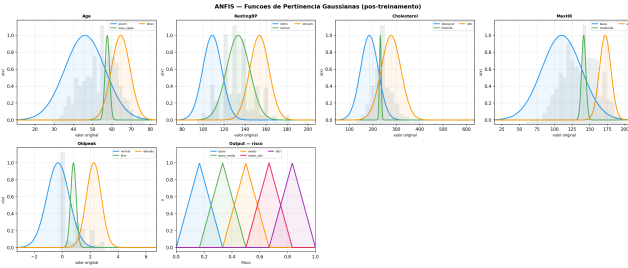


Fig. 1. Gaussian MFs learned by ANFIS after training (numerical features) and fixed triangular output MFs.

C. Confusion Matrices

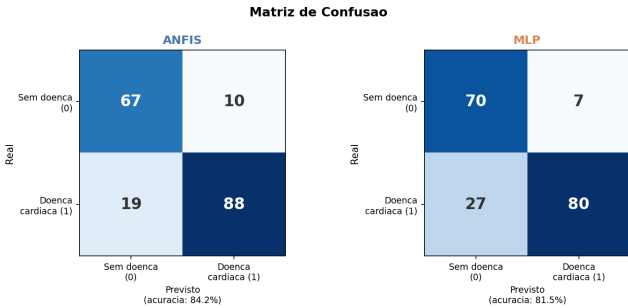


Fig. 2. Confusion matrices on the test set: ANFIS (left) and MLP (right).

ANFIS produces 18 fewer false negatives than the MLP on the test set. In clinical practice—where an undetected disease is the costlier error—this represents a meaningful diagnostic advantage and directly explains the higher F1-Score.

D. Extracted Linguistic Rules

Table IV lists the top-10 rules by mean firing strength. Two consistent patterns emerge:

- **High risk:** ChestPainType=ASY, low MaxHR, ExerciseAngina=Y, high Oldpeak, ST_Slope=Flat — established coronary ischemia markers [4].

- **Low risk:** ChestPainType=ATA, high MaxHR, ExerciseAngina=N, normal Oldpeak, ST_Slope=Up — consistent with normal exercise physiology.

#	Antecedent (abbreviated)	Risk
1	Age=young, CPT=ATA, MaxHR=high, Ang=N, OP=normal, ST=Up	LOW
2	Age=young, CPT=ASY, MaxHR=low, Ang=Y, OP=high, ST=Flat	HIGH
3	Age=old, CPT=ASY, ECG=ST, MaxHR=low, Ang=Y, OP=high, ST=Flat	HIGH
4	Age=young, CPT=ATA, RBP=high, MaxHR=low, Ang=N, OP=normal, ST=Up	LOW
5	Age=young, CPT=ATA, RBP=opt, MaxHR=low, Ang=N, OP=normal, ST=Up	LOW
6	Age=young, Sex=F, CPT=ATA, MaxHR=low, Ang=N, OP=normal, ST=Up	LOW
7	Age=old, CPT=ASY, ECG=L VH, MaxHR=low, Ang=Y, OP=high, ST=Flat	HIGH
8	Age=young, CPT=NAP, MaxHR=high, Ang=N, OP=normal, ST=Up	LOW
9	Age=young, CPT=ASY, FBS=1, MaxHR=low, Ang=N, OP=normal, ST=Flat	HIGH
10	Age=young, CPT=ASY, RBP=high, MaxHR=low, Ang=Y, OP=high, ST=Flat	HIGH

TABELA IV. Top-10 ANFIS rules by mean firing strength. CPT=ChestPainType, RBP=RestingBP, Ang=ExerciseAngina, OP=Oldpeak, FBS=FastingBS.

VII. CONCLUSION

ANFIS with mixed fuzzification and K-Means-generated rules achieves AUC-ROC 0.912 with only 228 parameters—13× fewer than the MLP—while producing 10 auditable linguistic rules that align with clinical literature.

The genuine interpretability obtained—Gaussian centers with physical meaning, per-rule TS consequents, and intelligible IF-THEN rules—differentiates ANFIS from post-hoc methods such as SHAP, where the explanation is generated after the fact.

Future work: Validate on larger and multi-class datasets; incorporate domain-expert knowledge in MF initialization; compare with classical ANFIS [2] and Mamdani systems; investigate formal MF definitions for categorical variables.

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